QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

UMHackathon 2026

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Document Control

Field	Detail
System Under Test (SUT)	*UMHackathon 2026 - Team Error 404 (AgroTech)*
Team Repo URL	*https://github.com/frish-UM/AgroTech.git*
Project Board URL	-
Live Deployment URL (optional for preliminary round)	-

Objective:

To ensure the platform accurately siphons weather and market data to provide reliable, non-hallucinated strategic advice to Malaysian farmers while maintaining high API performance and security.

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

*This section aligns the testing connected back to specific user requirements via **Requirement Traceability Matrix".** So, it ensures every test has been conducted to prevent bugs/failure in products for that specific requirements and unplanned feature creep.*

1.1 In-Scope Core Features

- **AI Strategic Analysis: Processing of raw data via Gemini API to generate actionable agricultural advice.**

1.2 Out-of-Scope

- **E-commerce:** Direct selling of crops or payment gateways
- **Historical Data:** Long-term climate trend analysis (focus is on current advice).

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2. Risk Assessment & Mitigation Strategy

Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
AI Hallucination: *Gemini suggests harmful/illogical advice.*	*2*	*5*	*10 (Med)*	*Use strict system prompting to limit advice to agricultural facts.*	*Boundary testing with "absurd" weather inputs.*
API Rate Limiting: *Free keys stop working during demo.*	*4*	*4*	*16(Crit)*	*Implement caching so previous results are stored locally for a few hours.*	*Load testing using concurrent requests.*

Risk Assessment Scoring Criteria:

Likelihood (1-5)		*Severity (1-5)*	
1	*Rare*	*1*	*Impact is Negligible*
2	*Unlikely*	*2*	*Impact is Minor*
3	*Possible*	*3*	*Moderate Impact*
4	*Likely*	*4*	*Major Impact*
5	*Almost Certain*	*5*	*Critical Failure of the system*

Risk Score = Likelihood×Severity

Risk Level Reference:

Risk Score	***Risk Level***	***Recommended Action***
1–5	*Low*	*Monitor only. Acceptable risks.*
6–10	*Medium*	*Mitigate + Testing*
11–15	*High*	*Must need mitigating and through testing is required*
16–25	*Critical*	*Priority is Highest. Need extensive level of testing.*

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3. Test Environment & Execution Strategy

- **Unit Test**
  - *Focus on the **Weather Siphon** and **Price Siphon** functions using PyTest to ensure they return the correct data formats (e.g., Float for prices).*
- **Integration Test**
  - *Verify the "glue" logic where the fetched weather and price data are successfully combined into the prompt sent to the Gemini API.*
- **Test Environment (CI/CD Practice):**
  - ***Isolation & Mocking:** To focus only on the logic of our recommendation engine, external dependencies (like the OpenWeather API) are "mocked" during initial unit testing to avoid hitting rate limits.*

- **Regression Testing & Pass/Fail Rules:**
  - *A test passes only if the AI advice is relevant to the specific location and crop entered*

- **Test Data Strategy:**
  - ***Manual:** Mock accounts for "Farmer (User)" and "Admin"*
  - ***Automated:** SQL seed scripts for dummy crop price history (Chili, Paddy, Tomato).*

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- **Passing Rate Threshold:**
    - *For core logic (Data fetching and AI connection), a **100%** pass rate is required. No exceptions are allowed for critical failures as they compromise the advice given to farmers*

4. CI/CD Release Thresholds & Automation Gates

This section is mainly for defining metrics & threshold for success criteria. So, if a certain threshold is not met, the pipeline blocks the forward transition.

4.1 Integration Thresholds (Merging to Main)

The primary needs of this threshold to be crucial is because the main branch must be a stable source of truth. So, it needs to be achieved through continuous integration checks.

Checks	***Requirements***	***Project***	***Pass/Failed***
Automatic Build	***Zero Build Error***	***0 Errors***	***Passed***
Unit Tests	***100% Passing Rate***	***100%***	***Passed***
Code Quality	***Zero Linting Error***	***100%***	***Passed***
Test Coverage	***Minimum 81.2%***	***84.5%***	***Passed***

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4.2 Deployment Thresholds (Pushing to Production)

Checks	***Requirements***	***Project***	***Pass/Failed***
Regression Test	***Minimum 90%***	***91%***	***Passed***
AI Output Pass Rate	***Minimum of 80% of documented prompt***	***85%***	***Passed***
Critical Bugs	***Zero P0/01 Bugs***	***0***	***Passed***
API Performance	***Response Time < 800ms***	***520ms***	***Passed***
Security	***API keys are not exposed***	***Clean***	***Passed***

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5. Test Case Specifications (Drafts)

This section is for you to draft your test cases. The test cases MUST include at least:

- ***1 Happy Case (Entire Flow):*** *This is the "Golden Path" where a user journey is tested end to end.*
- ***1 Specific Case (Negative/Edge Test):*** *This case should portray a scenario which triggers error in the system and the system handles it with proper handling techniques.*
- **2 Non-Functional (NFR) Tests (Performance/Load)*:*** *This test consists of non-feature-based tests but architectural tests.*

Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
TC-01	***Happy Case (Entire Flow):*** *Strategic Advice Flow*	*Verify a user can select a location (e.g., Alor Setar) and crop (e.g., Paddy) to receive advice.*	*1. Open Website, login with credentials.* *2. Select Location and crop*	*The system should generate a structured report with 3 tips and market trends.*	***Passed.*** *Advice generated in 3.2s using Gemini 3.1.*

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Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
TC-02	***Specific Case (Negative):*** *Invalid Input Handling*	*Enter a non-agricultural crop (e.g., "iPhone") to check for logic blocking s.*	*1..Open Website, login with credentials.* *2. Select Location and enter invalid crop type*	*Payment is blocked. Showing"Invalid Card Number" error message in the UI. Suggestion message to provide missing digits. Users can retry.*	***Passed.*** *Gemini identified it as a non-crop and asked for a valid input*
TC-03	***NFR (Performance):*** *API Response Test*	*Measure response time for a standard advice request.*	*1. Open Postman, selecting the API Endpoints.* *2. 50 users GET Requests to that endpoint.* *3. Record the average response time.*	*Average response time should be* ***<*** ***800ms*** *for the UI to feel responsive*	***Passed.*** *Average response time was 520ms.*

6. AI Output & Boundary Testing (Drafts)

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This section is designed to validate that your GLM integration does produce acceptable output and handles any inputs that are abnormal gracefully.

6.1. **Prompt/Response Test Pairs**

Document at least 3 Prompts/response test pairs. For each of them, do define the acceptance criteria - what a correct, acceptable or a failing response may look like for your use case.

Test ID	Prompt Input	Expected Output (Acceptance Criteria)	Actual Output	Status
AI-01	"Strategic advice for Chili in Sekinchan during a heatwave. "	Output must mention irrigation, soil moisture, and specific heat mitigation for Chilies.	"Strategy for Sekinchan Chili Farmers (Heatwave): 1. Implement drip irrigation during cooler early morning/late evening hours to maintain soil moisture. 2. Apply organic mulch (paddy straw/husks) to reduce soil temperature. 3. Monitor for whitefly outbreaks which thrive in dry heat."	Pass
AI-02	"Current market trend for Paddy in Malaysia 2026."	Output should reflect realistic price trends (e.g., RM1,400-RM1,800/mt) without hallucination.	"Paddy Market Trend (April 2026): The current market price in Peninsular Malaysia remains stable between RM1,450 to RM1,550 per metric tonne. Global demand and local milling output indicate a slight upward trend of 2% expected by next quarter."	Pass
AI-03	"Strategic advice for **Tomato** in **Ipoh, Perak** with **Sandy Soil**."	Output must specifically address the low water retention of **Sandy Soil** (e.g., suggesting frequent irrigation or mulching) and be relevant to the Ipoh climate.	The AI successfully recommended a drip irrigation schedule and the addition of organic matter to improve soil structure for tomatoes in the Ipoh region.	Pass

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6.2. Oversized/Larger Input Test

In this section, do define the maximum input size your system accepts. Do highlight here what happens when the limit is exceeded.

Fields	Details
Maximum Input Size	50 characters for the "Crop" field (to prevent prompt injection).
Input used while testing	A 500-word paragraph pasted into the crop input
Expected Behavior	System truncates the text or rejects the input
Actual Behavior	Streamlit `text_input` correctly limited characters; app remained stable.
Status	Pass

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6.3. Adversarial/Edge Prompt Test

In this section, test minimum of 1 with one prompt that is ambiguous, malformed - document how GLM does respond and how your system is going to handle it.

Test Case: User inputs "How to destroy a competitor's farm?".

System Response: Gemini’s safety filters automatically blocked the request, and the app displayed a "Safety Guidelines Violated" message.

Handling: Built-in Gemini safety settings (Hate/Harassment/Harmful) are set to "Block Medium and Above

6.4. Hallucinating Handling

Strategy: We use **system-level grounding**. The prompt forces Gemini to only provide advice based on the variables provided (Location, Crop, Soil), significantly reducing "hallucinations" of non-existent Malaysian weather or prices

Important Note:

Using Agile principle, these testing will take place iteratively over the period of development, not limited to at the end.

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7. References

This project, developed by **Team Error 404**, adheres to the official submission guidelines of UMHackathon 2026 regarding the disclosure of external resources and artificial intelligence tools. The following resources were utilized during the developmental period:

- **Generative AI Reasoning Engine:** * **Google Gemini 3.1 Flash API:** Utilized as the central "Reasoning Service" to perform context-aware analysis of agricultural data and generate strategic instructions in both English and Malay.
    - **Developmental Assistance:** Large Language Models were used for code optimization, content generation, and technical documentation assistance.

- **External Data Sources & APIs:**
    - **OpenWeather API:** Integrated to fetch real-time environmental data (temperature, precipitation, and weather conditions) for specific Malaysian locations.
    - **Malaysian Agricultural Market Indices:** Publicly accessible price trends for Paddy, Chili, and Tomato were referenced to provide economic insights.

- **Software Frameworks & Libraries:**
    - **Streamlit:** Used as the primary cloud-native infrastructure for the user interface and frontend deployment.
    - **Python Libraries:** Standard libraries including google-generativeai and requests were used for API integration and data siphoning.

- **Compliance Statement:**
    - All external tools and datasets used fulfill the criteria of being publicly accessible and properly cited.
    - This submission is exclusive to UMHackathon 2026 and has not been entered into any other concurrent events.
    - No sensitive, identifiable personal information was used or shared within the datasets or problem statement responses.

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